

Experiment 7

AIM: Browser Automation Testing using Selenium (Python)

→Install selenium and webdriver-manager

pip install selenium

pip install webdriver-manager

PS C:\Yash Ex7> pip install selenium

Defaulting to user installation because normal site-packages is not writeable

Collecting selenium

Downloading selenium-4.41.0-py3-none-any.whl.metadata (7.5 kB)

Collecting certifi>=2026.1.4 (from selenium)

Downloading certifi-2026.2.25-py3-none-any.whl.metadata (2.5 kB)

Collecting trio<1.0,>=0.31.0 (from selenium)

Downloading trio-0.33.0-py3-none-any.whl.metadata (8.5 kB)

Collecting trio-websocket<1.0,>=0.12.2 (from selenium)

Downloading trio-websocket-0.12.2-py3-none-any.whl.metadata (5.1 kB)

Collecting typing_extensions<5.0,>=4.15.0 (from selenium)

Downloading typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)Collecting
urllib3<3.0,>=2.6.3 (from urllib3[socks]<3.0,>=2.6.3->selenium)

Downloading urllib3-2.6.3-py3-none-any.whl.metadata (6.9 kB)

Collecting websocket-client<2.0,>=1.8.0 (from selenium)

Downloading websocket_client-1.9.0-py3-none-any.whl.metadata (8.3 kB)

Collecting attrs>=23.2.0 (from trio<1.0,>=0.31.0->selenium)

Downloading attrs-25.4.0-py3-none-any.whl.metadata (10 kB)

Collecting sortedcontainers (from trio<1.0,>=0.31.0->selenium)

Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl.metadata (10 kB)

Collecting idna (from trio<1.0,>=0.31.0->selenium)

Downloading idna-3.11-py3-none-any.whl.metadata (8.4 kB)

Collecting outcome (from trio<1.0,>=0.31.0->selenium)

Downloading outcome-1.3.0.post0-py2.py3-none-any.whl.metadata (2.6 kB)

Collecting sniffio>=1.3.0 (from trio<1.0,>=0.31.0->selenium)

Downloading sniffio-1.3.1-py3-none-any.whl.metadata (3.9 kB)

Collecting cffi>=1.14 (from trio<1.0,>=0.31.0->selenium)

Downloading cffi-2.0.0-cp313-cp313-win_amd64.whl.metadata (2.6 kB)

Collecting wsproto>=0.14 (from trio-websocket<1.0,>=0.12.2->selenium)

Downloading wsproto-1.3.2-py3-none-any.whl.metadata (5.2 kB)

Collecting pysocks!=1.5.7,<2.0,>=1.5.6 (from urllib3[socks]<3.0,>=2.6.3->selenium)

Downloading PySocks-1.7.1-py3-none-any.whl.metadata (13 kB)

Collecting pycparser (from cffi>=1.14->trio<1.0,>=0.31.0->selenium)

Downloading pycparser-3.0-py3-none-any.whl.metadata (8.2 kB)

Collecting h11<1,>=0.16.0 (from wsproto>=0.14->trio-websocket<1.0,>=0.12.2->selenium)

Downloading h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)

Downloading selenium-4.41.0-py3-none-any.whl (9.5 MB)

9.5/9.5 MB 458.9 kB/s 0:00:20Downloading

trio-0.33.0-py3-none-any.whl (510 kB)

Downloading trio-websocket-0.12.2-py3-none-any.whl (21 kB)

Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)

Downloading urllib3-2.6.3-py3-none-any.whl (131 kB)

Downloading PySocks-1.7.1-py3-none-any.whl (16 kB)

Downloading websocket_client-1.9.0-py3-none-any.whl (82 kB)

Downloading attrs-25.4.0-py3-none-any.whl (67 kB)

Downloading certifi-2026.2.25-py3-none-any.whl (153 kB)

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Downloading cffi-2.0.0-cp313-cp313-win_amd64.whl (183 kB)
Downloading outcome-1.3.0.post0-py2.py3-none-any.whl (10 kB)
Downloading sniffio-1.3.1-py3-none-any.whl (10 kB)
Downloading wsproto-1.3.2-py3-none-any.whl (24 kB)
Downloading h11-0.16.0-py3-none-any.whl (37 kB)
Downloading idna-3.11-py3-none-any.whl (71 kB)
Downloading pycparser-3.0-py3-none-any.whl (48 kB)
Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl (29 kB)
Installing collected packages: sortedcontainers, websocket-client, urllib3,
typing_extensions, sniffio, pysocks, pycparser, idna, h11, certifi, attrs, wsproto,
outcome, cffi, trio, trio-websocket, selenium
1/17 [websocket-client] WARNING: The
script wsdump.exe is installed in
'C:\Users\student\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2
kfra8p0\LocalCache\local-packages\Python313\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress
thSuccessfully installed attrs-25.4.0 certifi-2026.2.25 cffi-2.0.0 h11-0.16.0 idna-
3.11 outcome-1.3.0.post0 pycparser-3.0 pysocks-1.7.1 selenium-4.41.0 sniffio-1.3.1
sortedcontainers-2.4.0 trio-0.33.0 trio-websocket-0.12.2 typing_extensions-4.15.0
urllib3-2.6.3 websocket-client-1.9.0 wsproto-1.3.2

[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run:
C:\Users\student\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python
.3.13_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip
```

```
PS C:\Yash Ex7> pip install webdriver-manager
Defaulting to user installation because normal site-packages is not writeable
Collecting webdriver-manager
  Downloading webdriver_manager-4.0.2-py2.py3-none-any.whl.metadata (12 kB)
Collecting requests (from webdriver-manager)
  Downloading requests-2.32.5-py3-none-any.whl.metadata (4.9 kB)
Collecting python-dotenv (from webdriver-manager)
  Downloading python_dotenv-1.2.2-py3-none-any.whl.metadata (27 kB)
Collecting packaging (from webdriver-manager)
  Downloading packaging-26.0-py3-none-any.whl.metadata (3.3 kB)
Collecting charset_normalizer<4,>=2 (from requests->webdriver-manager)
  Downloading charset_normalizer-3.4.6-cp313-cp313-win_amd64.whl.metadata (41 kB)
Requirement already satisfied: idna<4,>=2.5 in
c:\users\student\appdata\local\packages\pythonsoftwarefoundation.python.3.13_qbz5n2k
fra8p0\localcache\local-packages\python313\site-packages (from requests->webdriver-
manager) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
c:\users\student\appdata\local\packages\pythonsoftwarefoundation.python.3.13_qbz5n2k
fra8p0\localcache\local-packages\python313\site-packages (from requests->webdriver-
manager) (2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in
c:\users\student\appdata\local\packages\pythonsoftwarefoundation.python.3.13_qbz5n2k
fra8p0\localcache\local-packages\python313\site-packages (from requests->webdriver-
manager) (2026.2.25)
Downloading webdriver_manager-4.0.2-py2.py3-none-any.whl (27 kB)
Downloading packaging-26.0-py3-none-any.whl (74 kB)
Downloading python_dotenv-1.2.2-py3-none-any.whl (22 kB)
Downloading requests-2.32.5-py3-none-any.whl (64 kB)
Downloading charset_normalizer-3.4.6-cp313-cp313-win_amd64.whl (154 kB)
Installing collected packages: python-dotenv, packaging, charset_normalizer,
```

```
requests, webdriver-manager
0/5 [python-dotenv] WARNING: The script
dotenv.exe is installed in
'C:\Users\student\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2
kfra8p0\LocalCache\local-packages\Python313\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning,
use --no-warn-script-location.
2/5 [charset_normalizer] WARNING: The
script normalizer.exe is installed in
'C:\Users\student\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2
kfra8p0\LocalCache\local-packages\Python313\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning,
use --no-warn-script-location.
Successfully installed charset_normalizer-3.4.6 packaging-26.0 python-dotenv-1.2.2
requests-2.32.5 webdriver-manager-4.0.2
[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run:
C:\Users\student\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python
.3.13_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip
```

selenium_login_test.py

```
# Step 2: Import Required Libraries [cite: 8]
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.chrome.service import Service
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from webdriver_manager.chrome import ChromeDriverManager
import time

# Step 3: Start Browser [cite: 14]
service = Service(ChromeDriverManager().install())
driver = webdriver.Chrome(service=service)
driver.get("https://the-internet.herokuapp.com/login")
driver.maximize_window()

# Step 4: Wait for Page Elements [cite: 19]
wait = WebDriverWait(driver, 10)

# Step 5: Enter Login Credentials [cite: 21]
username = wait.until(EC.presence_of_element_located((By.ID, "username")))
username.send_keys("tomsmith")
driver.find_element(By.ID, "password").send_keys("SuperSecretPassword!")

# Step 6: Click Login Button [cite: 26]
driver.find_element(By.CSS_SELECTOR, "button.radius").click()

# Step 7: Verify Login Success [cite: 28]
message = wait.until(EC.presence_of_element_located((By.ID, "flash")))
print("Login Message:", message.text)

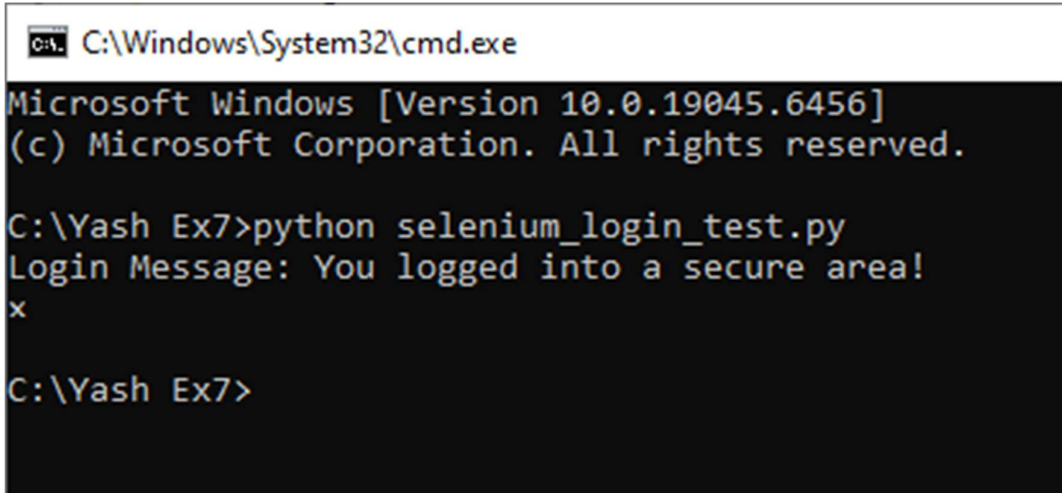
# Step 8: Logout [cite: 34]
driver.find_element(By.CSS_SELECTOR, "a.button.secondary.radius").click()

# Step 9: Close Browser [cite: 36]
time.sleep(3)
driver.quit()
```

CMD OUTPUT:

C:\Yash Ex7> python selenium_login_test.py
Login Message: You logged into a secure area!

x



```
C:\Windows\System32\cmd.exe

Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Yash Ex7>python selenium_login_test.py
Login Message: You logged into a secure area!
x

C:\Yash Ex7>
```

youtube_automate.py

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.keys import Keys # Added this to use keyboard keys
from selenium.webdriver.chrome.service import Service
from webdriver_manager.chrome import ChromeDriverManager
import time
```

```
service = Service(ChromeDriverManager().install())
driver = webdriver.Chrome(service=service)
```

try:

```
# 1. & 2. Open browser and navigate to YouTube
driver.get("https://www.youtube.com")

# 3. Wait for 2 seconds to allow the page to load
time.sleep(2)

# 4. Maximize the browser window
driver.maximize_window()

# 5. Locate the YouTube search box
search_box = driver.find_element(By.NAME, "search_query")

# 6. Enter the text "open source software" in the search field
search_box.send_keys("open source software")

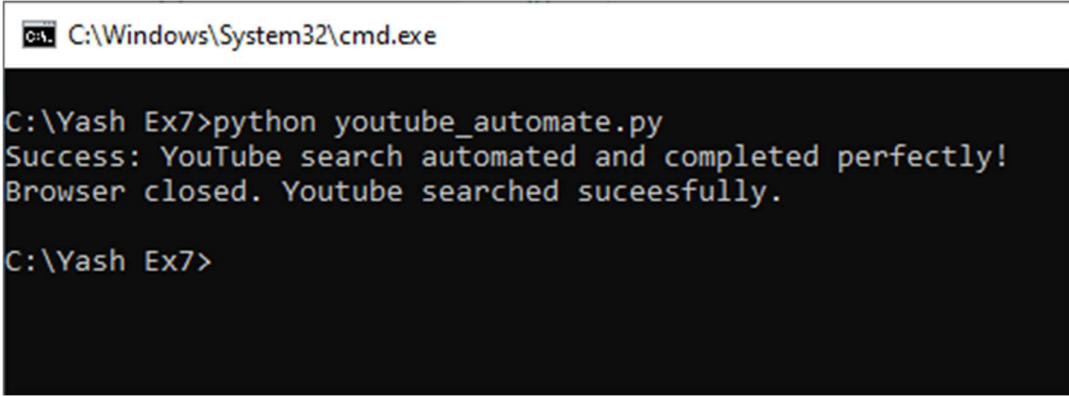
# 7. Press the ENTER key to search (Replacing the click button logic)
search_box.send_keys(Keys.ENTER)

# 8. Wait for 5 seconds to view the search results
time.sleep(5)
```

```
# ---> NEW: Print a success message to the terminal <---  
print("Success: YouTube search automated and completed perfectly!")  
  
finally:  
    # 9. Close the browser  
    driver.quit()  
  
# ---> NEW: Confirm the browser closed <---  
print("Browser closed. Youtube searched suceesfully.")
```

CMD OUTPUT:

```
C:\Yash Ex7>python youtube_automate.py  
Success: YouTube search automated and completed perfectly!  
Browser closed. Youtube searched suceesfully.
```



The screenshot shows a Windows Command Prompt window with the title bar "C:\Windows\System32\cmd.exe". The command prompt displays the following text:

```
C:\Yash Ex7>python youtube_automate.py  
Success: YouTube search automated and completed perfectly!  
Browser closed. Youtube searched suceesfully.  
  
C:\Yash Ex7>
```